

Jonathan J. Froehlich

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Experimental-computational scientist who builds scalable perturbation and functional-genomics systems to answer complex biological questions.

PROFESSIONAL EXPERIENCE

[\[LinkedIn\]](#)

01/2023–07/2026

Postdoctoral researcher

Max Planck Institute for Molecular Genetics, Berlin, DE, Systems Epigenetics, Dr. Edda Schulz.

- Integrated quantitative Oct4 CRISPRi perturbations with scRNA-seq and ATAC-seq to identify dose-responsive gene programs and candidate regulatory DNA in mouse embryonic stem cells.
- Established experimental and computational workflows spanning scRNA-seq, ATAC-seq, and MPRA to prioritize and test cis-regulatory elements across transcription-factor doses.

01/2014–12/2021

Doctoral researcher

Max Delbrueck Center for Molecular Medicine & Berlin Inst. for Medical Systems Biol., Berlin, DE, Systems Biology of Regulatory Elements, Prof. Nikolaus Rajewsky

- Developed an *in vivo*, parallelized CRISPR-Cas9 screening in *C. elegans* to test >1,000 gene regulatory mutations, a 100-fold increase over previous approaches.
- Designed a kilobase-scale amplicon-sequencing protocol and partnered with a computational collaborator to build specialized analysis software for high-throughput regulatory screens.
- Discovered that adjacent microRNA binding sites in the *lin-41* 3'-UTR show independent and compensatory regulatory effects.

04/2012–09/2012

01/2010–09/2010

Bachelor's & Master's student

Max Planck Institute of Immunobiology and Epigenetics, Freiburg, DE, Epigenetics, Prof. Thomas Jenuwein

- Developed a semi-automated microscopy and image-analysis workflow for quantifying heterochromatin state in reporter cells.
- Established a microscopy-based screen with a kinase inhibitor library to demonstrate that the PI3K signaling pathway promotes cellular heterochromatin condensation.

EDUCATION

2014–2022 **Dr. rer. nat., Molecular Biology**, Humboldt University, Berlin, DE
2010–2012 **MSc, Biology**, Ludwig Maximilian University, Munich, DE
2007–2010 **BSc, Molecular Biotechnology**, FH Campus Vienna, Vienna, AT

TECHNICAL EXPERIENCE

LABORATORY

2023–2026	Tunable CRISPRi, ATAC-seq, scRNA-seq, MPRA
2023–2026	Cell culture of embryonic stem cells
2015–2026	Synthetic DNA construct design, cloning, delivery, line generation
2009–2026	Epifluorescence and confocal microscopy; flow cytometry
2007–2026	Molecular biology of RNA, DNA, and proteins
2014–2022	Multiplexed CRISPR-Cas9 screening of noncoding DNA in <i>C. elegans</i>
2014–2022	<i>C. elegans</i> handling and phenotype screening
2017–2021	mRNA/DNA amplicon sequencing Illumina & PacBio
2010, 2012	96-well semi-automated microscopy-based compound screening
2011–2012	<i>D. melanogaster</i> handling and maintenance
2009–2011	Cell culture of fibroblast cell lines and embryonic stem cells

COMPUTATIONAL

2024–2026	ATAC-seq and motif-grammar analysis; ChromBPNet; TF motifs
2023–2026	scRNA-seq analysis with STARsolo, R, and Bioconductor
2022–2026	Reproducible analysis of RT-qPCR, FACS, and other data with R
2014–2026	Collaborated in interdisciplinary experimental-computational teams
2007–2026	Adobe Illustrator/Photoshop, Geneious/ApE/BioEdit
2010, 2012	Automated image analysis with CellProfiler, and CPython/SQL
2008, 2020	3D imaging: Amira for segmentation, Blender to measure volumes

PUBLICATIONS

[Google Scholar] [ORCID]

- 2025 Till Schwämmle, Gemma Noviello, Eleni Kanata, **Jonathan J Froehlich**, Melissa Bothe, Aybuge Altay, Jade Scouarnec, Vivi-Yun Feng, Martin Vingron, Edda G Schulz, “Reporter CRISPR screens decipher cis- and trans-regulatory principles at the Xist locus”, *Nature Structural & Molecular Biology*, [DOI].
- 2023 **Jonathan J Froehlich**, Nikolaus Rajewsky, “*C. elegans* gene regulatory alleles and reporter bashing studies”, *Micropublication Biology*, [DOI].
- 2022 **Jonathan J Froehlich**, “Parallel Genetics of Gene Regulatory Sequences in *Caenorhabditis elegans*”, Dissertation, Humboldt University Berlin, [DOI].
- 2021 **Jonathan J Froehlich***, Bora Uyar*, Margareta Herzog, Kathrin Theil, Petar Glažar, Altuna Akalin, Nikolaus Rajewsky, “Parallel genetics of regulatory sequences using scalable genome editing in vivo”, *Cell Reports*, [DOI] [news] [software] *co-first
- 2021 **Jonathan J Froehlich**, Nikolaus Rajewsky, Collin Ewald, “Estimation of *C. elegans* cell- and tissue volumes”, *Micropublication Biology*, [DOI].
- 2013 Sylvain Maenner, Marisa Müller, **Jonathan Froehlich**, Diana Langer, Peter B Becker, “ATP-dependent roX RNA remodeling by the helicase maleless enables specific association of MSL proteins”, *Molecular Cell*, [DOI].
- 2012 Silvia Dambacher, Wen Deng, Matthias Hahn, Dennis Sadic, **Jonathan Froehlich**, Alexander Nuber, Christian Hoischen, Stephan Diekmann, Heinrich Leonhardt, Gunnar Schotta, “CENP-C facilitates the recruitment of M18BP1 to centromeric chromatin”, *Nucleus*, [DOI].

ACTIVITIES

LEADERSHIP & ENGAGEMENT

2023–2026	Postdoc association representative
2023–2026	Maintainer of team's website
2023–2026	Led one main- and collaborated on five side projects
2015, 2023, 2025	Co-organized three research retreats for 15-30 participants
2023	Co-hosted a distinguished speaker at the institute
2014–2023	Contributed 17 biological diagrams to Wikipedia [Wikipedia contributions]
2023	Designed the program booklet and supported conference organization
2014–2022	Led one main- and two side projects from conceptualization to publication
2020–2021	Coordinated weekly team meetings, established guidelines
2020–2021	Initiated a community gardening project on the institute's terrace
2016–2021	Coordinated monthly social event "After Hour" at the institute, MDC/BIMSB
2014–2019	Communicated science to public at 4 events "Long Night of Science Berlin"

SIDE PROJECTS

2026	[Papers-to-table] extracts information from scientific publications
2026	[Curated resources] for research work, communication, lab culture
2025	[Game] that explores limitations of citation counts

TEACHING

04/2024–09/2025	Supervised Master's student, CRISPRi lines and cis-regulatory DNA reporter
06/2018–07/2018	Supervised Master's student, knock-in C. elegans
02/2023	Instructor at university course on CRISPR, Free University, Berlin
03/2014	Instructor at university course on C. elegans, Humboldt University, Berlin

SELECTED PRESENTATIONS

2025	Institute Seminar, Max Planck Institute for Molecular Genetics, Berlin
2019	Selected Talk, 22nd International C. elegans Conference, Los Angeles
2017	Selected Talk, 21st International C. elegans Conference, Los Angeles
2016	Poster, EMBO "Complex Life of mRNA", Heidelberg
2015/2017/2019	Talks, Student Seminar Series, Max Delbrueck Center, Berlin